

Fourier Polynomials

from Student Research Projects in Calculus

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May 12, 2026

This problem was found in the book *Student Research Projects in Calculus* written by Marcus Cohen, Edward D. Guaghan, Arthur Knowbel, Douglas S. Kurtz, and David Penegelly. The problem is named 'Fourier Polynomials'. Below is my proof for the problems presented.

Let f be a continuous function on the interval $[-\pi, \pi]$. We define the Fourier coefficients of f by the formulae:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) \, dx$$

and

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) \, dx$$

for $n = 0, 1, 2 \dots$ ($n \in \mathbb{N}_0$).

We also define Fourier polynomials of f to be:

$$P_N(x) = \frac{a_0}{2} + \sum_{n=1}^N (a_n \cos(nx) + b_n \sin(nx))$$

Now we begin the problems deemed as 'an introduction to some of Fourier's work'.

- (a) Let $f(x) = x$. Find all of the Fourier coefficients of f . What are the first five Fourier polynomials P_N associated to f ? Plot the function f and these five polynomials.

Proof. First we let $a = 0$. This will give us:

$$\frac{1}{\pi} \int_{-\pi}^{\pi} x \, dx$$

which will evaluate to 0 using basic integration.

If we let $g(x) = x \cos(nx)$, we can show that $g(x)$ is odd by plugging in $-x$ for x :

$$g(-x) = (-x) \cos(-nx) \tag{1}$$

$$= -x \cos(nx) \tag{2}$$

$$= -g(x) \tag{3}$$

By proving that $g(x)$ is an odd function, we can conclude that any integral of reflective bounds (in this case, $[-\pi, \pi]$) will be evaluated to 0.

$$\therefore f(x) = x \implies a_n = 0 \forall n$$

This will simplify our Fourier polynomial definition to:

$$P_N(x) = \sum_{n=1}^N b_n \sin(nx)$$

When evaluating $b_1 - b_6$ using integration by parts, we are given:

$$\begin{aligned} b_1 &= 2 & b_2 &= -1 & b_3 &= \frac{2}{3} \\ b_4 &= -\frac{1}{2} & b_5 &= \frac{2}{5} & b_6 &= -\frac{1}{3} \end{aligned}$$

We can notice two patterns for the terms of $2n$ and $2n + 1$ as follows:

$$b_{2n+1} = \frac{2}{2n+1} \quad b_{2n} = -\frac{1}{n}$$

However, we can find a more concise pattern for both even and odd cases to simplify the formula. Multiplying the numerators and denominators gives

$$b_n = \frac{-2\pi(-1)^n}{\pi n}$$

Since π appears in both the numerator and denominator, we can simplify:

$$b_n = \frac{-2(-1)^n}{n}$$

Notice that multiplying $(-1)^n$ by -1 increases the exponent by one:

$$-(-1)^n = (-1)^{n+1}$$

Therefore,

$$b_n = \frac{2(-1)^{n+1}}{n}$$

Thus, the Fourier coefficients for $f(x) = x$ are

$$a_n = 0$$

and

$$b_n = \frac{2(-1)^{n+1}}{n}$$

for all $n \geq 1$.

Our final part of the question asks for the first five Fourier polynomials of $P_N(x)$ along with their graphs on the interval $[-\pi, \pi]$

$$P_1(x) = 2 \sin x$$

$$P_2(x) = 2 \sin x - \sin 2x$$

$$P_3(x) = 2 \sin x - \sin 2x + \frac{2}{3} \sin 3x$$

$$P_4(x) = 2 \sin x - \sin 2x + \frac{2}{3} \sin 3x - \frac{1}{2} \sin 4x$$

$$P_5(x) = 2 \sin x - \sin 2x + \frac{2}{3} \sin 3x - \frac{1}{2} \sin 4x + \frac{2}{5} \sin 5x$$

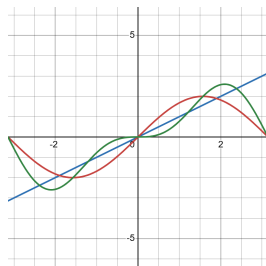


Figure 1: $x, P_1(x), P_2(x)$

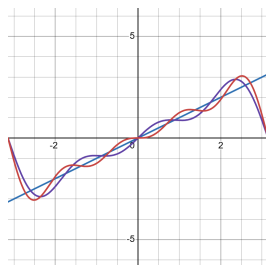


Figure 2: $x, P_3(x), P_4(x)$

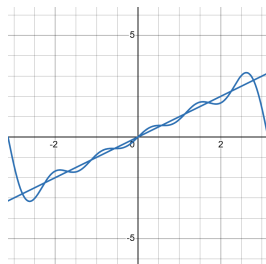


Figure 3: $x, P_5(x)$

We can see from our graphs that there is an obvious trend of accuracy in approximations. $\square$

(b) Let f be an odd function, continuous on the interval $[-c, c]$. Prove that

$$\int_{-c}^c f(x) \, dx = 0$$

Proof.

$$I = \int_{-c}^c f(x) \, dx = \int_0^c f(x) \, dx + \int_{-c}^0 f(x) \, dx$$

Let

$$I_2 = \int_{-c}^0 f(x) \, dx$$

Using a u-substitution, we can let $u = -x$, so $du = -dx$. This gives us:

$$I_2 = \int_c^0 f(-u) (-du)$$

Since $f(x)$ is an odd function, we know that $f(-x) = -f(x)$. With this information, we can simplify our integral.

$$\int_c^0 f(u) \, du = - \int_0^c f(u) \, du$$

u is what is called a 'dummy variable', so with a re-substitution of our original x and plugging I_2 back into the original integral, we get:

$$\begin{aligned} I &= \int_0^c f(x) \, dx + \left(- \int_0^c f(x) \, dx \right) \\ &= 0 \end{aligned}$$

$$\therefore \int_{-c}^c f(x) \, dx = 0$$

□

- (c) In part (a), something special happened because the function, $f(x) = x$, is an odd function on $[-\pi, \pi]$. What pattern did you notice? Next, let f be any odd function on $[-\pi, \pi]$. Make a conjecture about the Fourier coefficients of f . Prove your conjecture.

Conjecture 1. *For any odd function $f(x)$, the a_n Fourier coefficients will be 0.*

Proof. We compute

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

where $f(x)$ is odd and $\cos(nx)$ is even.

Let

$$g(x) = f(x) \cos(nx)$$

Then

$$g(-x) = f(-x) \cos(-nx)$$

Since $f(x)$ is odd and $\cos(nx)$ is even,

$$f(-x) = -f(x), \quad \cos(-nx) = \cos(nx)$$

Thus

$$g(-x) = -f(x) \cos(nx) = -g(x)$$

So $g(x)$ is an odd function.

Therefore,

$$\int_{-\pi}^{\pi} g(x) dx = 0 \quad \Rightarrow \quad a_n = 0$$

$\forall n \geq 1$.

Hence, all cosine coefficients vanish for any odd function. $\square$

- (d) Another special collection of functions on an interval $[-c, c]$ is the even functions. After thinking about what happens for the odd functions, make a conjecture about the Fourier coefficients of an even function. Let $f(x) = x^2$; check your conjecture using this even function. Explain why your conjecture is true for any function.

For this part, we will combine a similar approach to what we have done in both parts (a) and (c).

Conjecture 2. *For any even function $f(x)$, the b_n Fourier coefficients will be 0 and a_n coefficients will $\neq 0$.*

Proof. Recall that

$$P_N(x) = \frac{a_0}{2} + \sum_{n=1}^N a_n \cos(nx) + b_n \sin(nx)$$

We will first evaluating a_n coefficients of a_0 through a_5 .

$$\begin{aligned} a_0 &= \frac{2\pi^2}{3} & a_1 &= -4 & a_2 &= 1 \\ a_3 &= -\frac{4}{9} & a_4 &= \frac{1}{4} & a_5 &= -\frac{4}{25} \end{aligned}$$

Notice that the term $(-1)^n$ alternates signs depending on whether n is even or odd:

$$(-1)^n = \begin{cases} 1 & \text{if } n \text{ is even} \\ -1 & \text{if } n \text{ is odd} \end{cases}$$

Now consider the formula

$$a_n = \frac{4(-1)^n}{n^2}$$

We verify this formula for both even and odd cases.

If n is even, then $(-1)^n = 1$, so

$$a_n = \frac{4(1)}{n^2} = \frac{4}{n^2}$$

For example,

$$a_2 = \frac{4}{2^2} = 1$$

and

$$a_4 = \frac{4}{4^2} = \frac{1}{4}$$

which agrees with the even coefficients above.

If n is odd, then $(-1)^n = -1$, so

$$a_n = \frac{4(-1)}{n^2} = -\frac{4}{n^2}$$

For example,

$$a_1 = -\frac{4}{1^2} = -4$$

and

$$a_3 = -\frac{4}{3^2} = -\frac{4}{9}$$

which agrees with the odd coefficients above.

Therefore, both cases may be summarized by the compact formula

$$a_n = \frac{4(-1)^n}{n^2}$$

for all $n \geq 1$.

As we found a_0 , we can simplify $P_N(x)$

$$P_N(x) = \frac{\pi^2}{3} + \sum_{n=1}^N a_n \cos(nx) + b_n \sin(nx)$$

Since $f(x)$ is even and $\sin(nx)$ is odd, their product $f(x) \sin(nx)$ is an odd function. Therefore,

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = 0.$$

Using this information, we can make a broader claim to the conjecture we made earlier that any even function will allow b_n to always $= 0$.

If we let $f(x)$ be any even function, our b_n coefficient is defined as:

$$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) \, dx$$

Let

$$g(x) = f(x) \sin(nx) \tag{4}$$

$$g(-x) = f(-x) \sin(-nx) \tag{5}$$

This is almost an exact opposite case of what we saw in part (c), rather $f(x)$ is defined as an even function, and $\sin x$ is a known odd function.

$$= -f(x) \sin(nx) = -g(x) \tag{6}$$

This proves that $g(x)$ is an odd function. Using what we learned from part (b), we can determine that every b_n coefficient will evaluate to 0 when $f(x)$ is an even function.

□